Nonnegative and positive factorizations

Complexity of globally optimal nonnegative rank-two approximation

Difficulty: challenging **Importance:** interesting to the community

Status: Partially resolved

Rating rationale: Challenging because rank two still requires a global complexity classification despite tractable special inputs; community importance comes from the basic gap between SVD approximation and constrained NMF.
Topic: low-rank approximation; computational complexity

Problem statement

Given an arbitrary $X \in \mathbb{Q}_+^{m\times n}$ and $\tau \in \mathbb{Q}_{\geq 0}$, determine the complexity of deciding whether

$$\exists W \in \mathbb{R}_+^{m\times 2},\, H \in \mathbb{R}_+^{2\times n} : \quad \sum_{i=1}^{m}\sum_{j=1}^{n}\left(X_{ij} - \sum_{k=1}^{2} W_{ik}H_{kj}\right)^2 \leq \tau.$$

In particular, is there a deterministic algorithm polynomial in the total binary input length, or is this decision problem NP-hard under polynomial-time many-one reductions? This is a complexity-classification question, without an assumption that these two outcomes exhaust the possibilities. The factors may have real entries; only the data and threshold must be rational. Their inner dimension is at most two, with a zero factor column permitted.

This fixes an exact decision interpretation of the literature's global rank-two NMF optimization question. There is no promise that X itself has rank two. When a rank-two truncated SVD of X is nonnegative, a best nonnegative rank-two approximation is obtainable from it. Arbitrary input can fall outside this tractable special case, and alternating nonnegative least squares need not find a global optimum.

References

Gillis, *Nonnegative Matrix Factorization*, §6.1.3, p. 199, Theorem 6.6 and final paragraph. Lindy, Noferini, and Van Dooren, *On rank-2 Nonnegative Matrix Factorizations and their variants*, §1, with §3's suboptimal approximation and §4's ANLS initialization.

Status check — 2026-09-10

Rechecked the Lindy–Noferini–Van Dooren preprint and Noferini's December 2025 seminar, which explicitly retains the rank-two complexity question. Inputs having a nonnegative best rank-two SVD approximation form a proved tractable subclass. Searches also found Gouveia–Wiebe's 2026 integer-factor problem; its integer factors and exact-rank input are different. The 2026 constrained Gram-feasibility hardness result also has additional affine entry constraints and is not this unconstrained approximation problem. No unrestricted rational-input decision classification was located.